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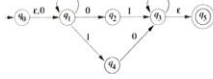
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Theory Of Computation [BITE306L] CAT-1 C2 2024-2025

Class Number(s) : Harshita Patel
Date of Examination : VL2024250503403, 3396, 3409
Exam Duration : 29/01/2025
Maximum Marks: 50
General instruction(s):

- Answer All Questions
- M - Max mark, CO - Course Outcome, BL - Blooms Taxonomy Level (1 - Remember, 2 - Understand, 3 - Apply, 4 - Analyse, 5 - Evaluate, 6 - Create)
- Course Outcomes (Type the CO statements covered in this question paper. Use the CO number as per the syllabus copy)
 - Analyze the deterministic finite machine and non-deterministic finite automata to accept the languages.
 - Use and apply important properties of finite automata to derive regular expressions from finite automata and vice versa.

Q. No.	Question	M	CO	BL
2.	Construct a Minimal DFA that accepts all strings of a's & b's for the following languages: i. $L = \{a^n b^m \mid n \geq 2, m \geq 1\}$ ii. $L = \{w \mid n(w) \bmod 3 = 1\}$ Also, demonstrate the DFA's acceptance or rejection by few valid and invalid strings.	5+5	1	6
2.	Construct an NFA that is equivalent to the given finite automaton. Determine the regular language accepted by the NFA. Finally, provide a neatly formatted transition table for the constructed NFA. 	10	1	6
3.	How do unreachable states affect the minimization process? Given a DFA, apply the minimization algorithm to construct an equivalent DFA with the minimum number of states.	2+8	1	3

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Theory Of Computation [BITE306L] CAT-1 C2 2024-2025

VIT Vellore Institute of Technology SCHOOL OF COMPUTER SCIENCE ENGINEERING & INFORMATION SYSTEMS CONTINUOUS ASSESSMENT TEST - I WINTER SEMESTER 2024-2025 Slot: C2+TC2+TCC2				
4.	Differentiate between finite automata with output and finite automata without output. Design a Moore machine for a binary input sequence such that if it has a substring 101, the machine output A, if the input has substring 110, it outputs B otherwise it outputs C.	2+8	1	3
5.	Describe the relationship between regular expressions, deterministic finite automata (DFAs), and non-deterministic finite automata (NFAs) in the context of formal language theory. Find the regular expression for the following regular languages: a. All the string of 0's & 1's where the 2 nd symbol from the right end of the string is always 0. b. All the string which should have at least one 0 and at least one 1.	4+3+3	2	3

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Theory Of Computation [BITE306L] CAT-1 C1 2024-2025

Maheshwari G

Exam Duration: 90 Min.

Maximum Marks: 50

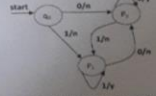
1. a) Differentiate between DFA and NFA with their State transition diagram. [5M]
b) Construct DFA to recognize odd number of 1's and even number of 0's? [5M]
2. a) Given that L_1 and L_2 are regular languages, is $L_1 \cup L_2$ also a regular language. Justify. [2.5M]
b) If L is a regular language over alphabet Σ , then complementation of L is $\bar{L} = \Sigma^* - L$. Explain with reasons. [2.5M]
c) Given that L_1 and L_2 are regular languages, is $L_1 \cap L_2$ also a regular language. Justify. [2.5M]
d) If $L_1 = \{a, b\}$ and $L_2 = \{aa, bb\}$, what would be the resulting concatenated language $L = L_1 \cdot L_2$? [2.5M]
3. Deduce the following e-NFA to DFA. [10M]

states	ϵ	a	b	c
$\rightarrow p$	$\{q, r\}$	ϕ	$\{q\}$	$\{q, r\}$
q	ϕ	$\{p\}$	$\{r\}$	$\{q\}$
r	ϕ	ϕ	ϕ	$\{r\}$

states	0	1
$\rightarrow q_0$	q_1	q_1
q_1	q_2	q_4
q_2	q_3	q_2
q_3	q_4	q_5
q_4	q_3	q_4
q_5	q_4	q_5
q_6	q_6	q_1

Starting State: q_0 and Final State: $\{q_3, q_4, q_5\}$.

5. a) Convert the following Mealy machine to an equivalent Moore machine. [5M]



b) Design a Mealy Machine accepting the language consisting of Strings from Σ^* , where $\Sigma = \{a, b\}$ and the string should end with either 'aa' or 'bb'. The given output alphabet is $\bar{A} = \{0, 1\}$. [5M]



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Theory Of Computation [BITE306L] FAT C2 2025-2026



Final Assessment Test - April 2025

Course: BITE306L

Theory of Computation

Class: BITE306L

Class: BITE306L

Time: Three Hours

Time: Three Hours

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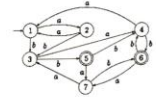
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Answer ALL Questions

(10 x 10 = 100 Marks)

1. Can a DFA accept an infinite language? Find the minimal DFA for the following regular languages:
a) $L = \{a^n b a^m \mid n \in \mathbb{N}, m \in \mathbb{N}\}$ [5]
b) $L = \{a^n b a^m \mid n \in \mathbb{N}, m \in \mathbb{N}\}$ [5]

2. Why do we remove unreachable states before starting the minimization process? Consider the following DFA:



3. Minimize this DFA and provide the transition table for the minimized DFA.

4. What are the key advantages of Moore machines over Mealy machines? Design a Moore machine that scans a sequence of 0s and 1s and generates output 'X' if the input string terminates in '00', output 'Y' if the string terminates in '11', and output 'C' otherwise. For instance, for the input '010011', find the corresponding output string.

5. Construct a Finite Automata that accepts the language described by the given regular expression.

- a) $a^*b^*c^*$ [2.5]
- b) $(a+b)^*ab^*b^*$ [2.5]
- c) $a^*b^*c^*a^*$ [2.5]
- d) $a^*(a+b)^*a^*$ [2.5]

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Theory Of Computation [BITE306L] FAT C1 2025-2026

Time: Three Hours

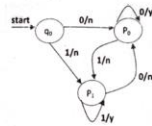
Max. Marks: 100

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Answer ALL Questions

(10 X 10 = 100 Marks)

1. a) Construct a DFA, which accepts those words from $\Sigma = \{a, b\}$ such that the number of a's is even and the number of b's is divisible by three. [5]
- b) Construct a DFA to recognize odd number of 1's and even number of 0's over $\Sigma = \{0, 1\}$. [5]
2. Define regular and non-regular language. Prove using pumping lemma, the following language is not regular language $L = \{w \in \{0, 1\}^* \mid w \text{ contains more 0's than 1's}\}$.
3. Convert the following Mealy machine to an equivalent Moore machine.



4. Derive the CFG for the given Context Free Language
 $L = \{v^r t^s \mid r, s \geq 0\}$ and check the ambiguity of the given grammar.
5. Construct the given Context Free Grammar $G = (V, T, P, S)$ into Chomsky normal form where P is the start symbol.
- $P \rightarrow RPT$
 $Q \rightarrow RP/Ra$
 $R \rightarrow bRP/Rb/c$
 $T \rightarrow PaP/R/aa$



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